

FinPredict

Intelligent Loan Approval System

User Manual

For Non-Technical Users

Version 1.0 | April 2026

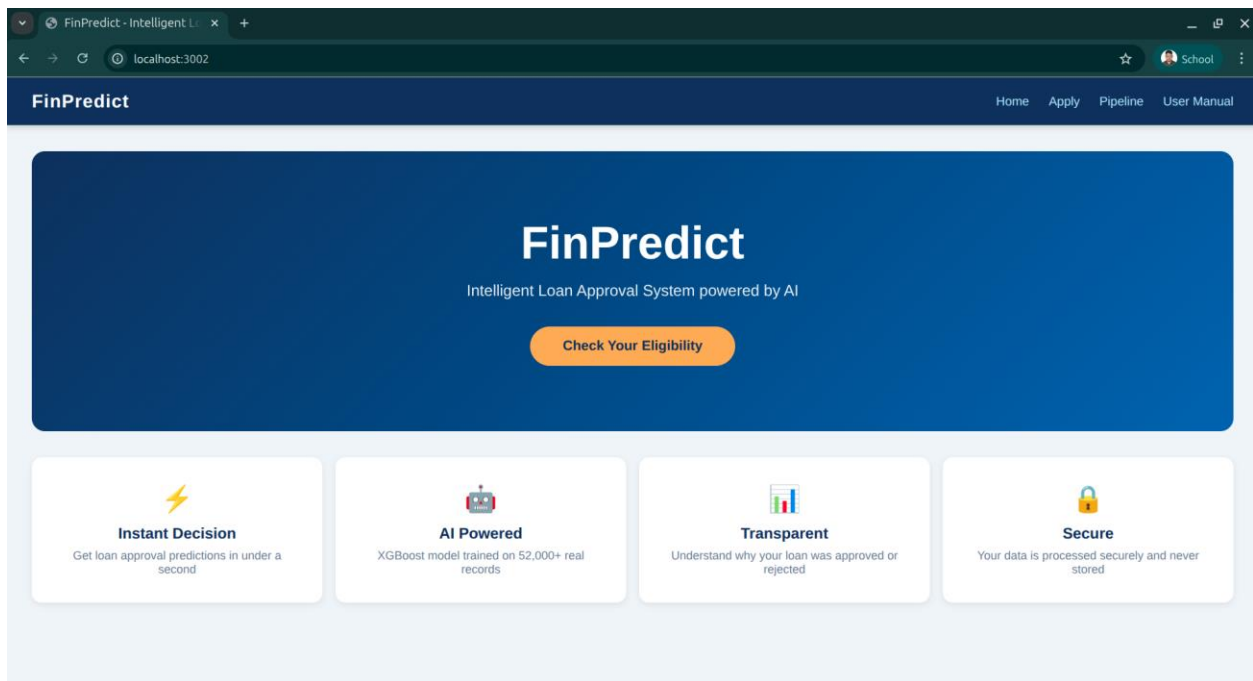
1. What is FinPredict?

FinPredict is an AI-powered web application that instantly tells you whether your loan application is likely to be approved or rejected by a financial institution. Instead of waiting days for a loan officer to review your application, FinPredict gives you an answer in seconds using artificial intelligence.

You simply fill in a form with your personal and financial details, click a button, and the system immediately shows you:

- Whether your loan would be Approved or Rejected
- How confident the system is in its prediction (approval probability)
- Your risk level — Low, Medium, or High
- A personalised recommendation to improve your chances

Note: FinPredict is a prediction tool to help you understand your chances before formally applying. The final loan decision is always made by the financial institution.



2. Getting Started

2.1 How to Open the Application

FinPredict runs in your web browser — you do not need to install any software. Simply open your browser (Chrome, Firefox, or Edge) and go to the address provided by your system administrator.

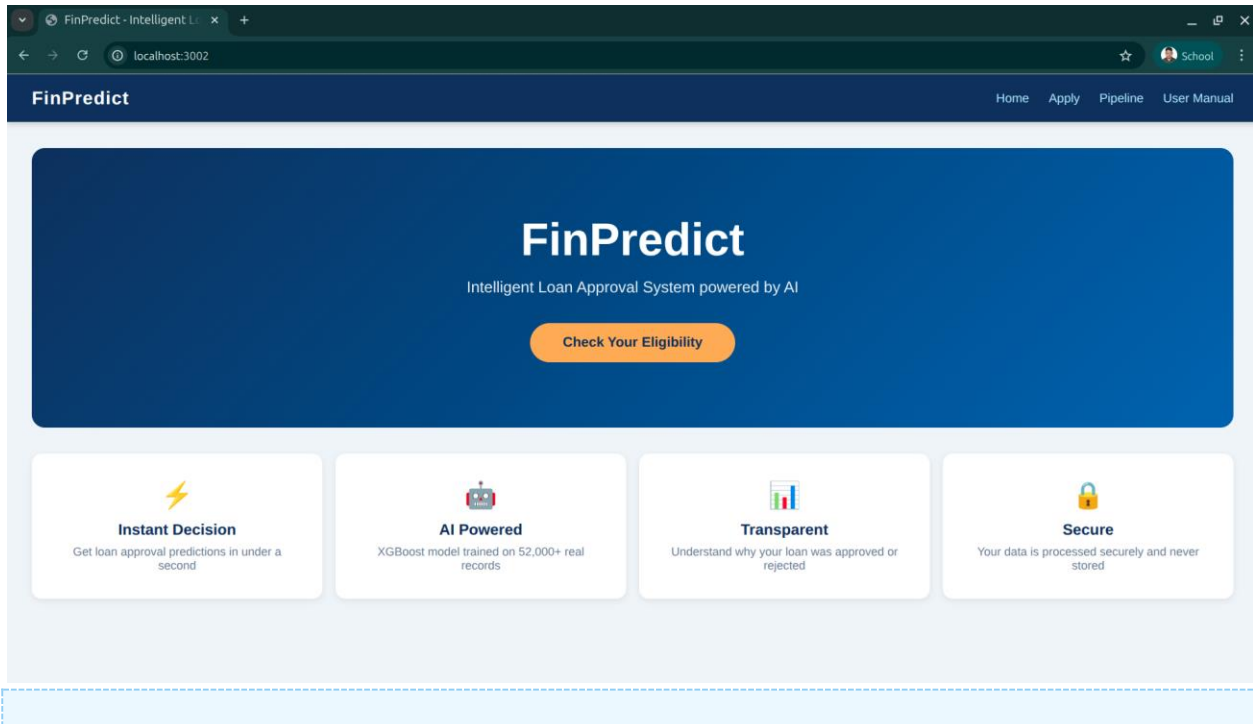
Item	Details
Application Address	http://localhost:3002 (or as provided by your administrator)
Browser Support	Google Chrome, Mozilla Firefox, Microsoft Edge
Internet Required	No — works on your local network
Login Required	No — open access

 **Tip:** For the best experience, use Google Chrome and keep your browser window maximised.

2.2 Navigating the Application

At the top of every page you will see the navigation bar with four links:

Link	What It Does
Home	Returns you to the welcome page
Apply	Opens the loan eligibility check form — this is where you enter your details
Pipeline	Shows the technical data processing pipeline (for administrators)
User Manual	Opens this user manual inside the application



3. Checking Your Loan Eligibility

The main feature of FinPredict is the loan eligibility check. Follow the steps below carefully.

Step 1 — Go to the Apply Page

Click the Apply button in the navigation bar at the top, or click the large Check Your Eligibility button on the home page. You will be taken to the Loan Application Form.

Step 2 — Fill in Personal Details

The first section asks for your personal information. Fill in each field:

Field	What to Enter	Example
Gender	Select Male or Female from the dropdown	Male
Age	Enter your age in years (must be 18 or above)	35
Marital Status	Select Married, Single, or Divorced	Married
Dependents	Number of people financially dependent on you (e.g. children)	2
Education	Your highest level of education completed	Graduate

Loan Eligibility Check
Fill in your details below to instantly check if your loan will be approved.

Loan Application Form

Personal Details

Gender: Male

Age: e.g. 30

Marital Status: Married

Dependents: e.g. 2

Education: Graduate

Employment Details

Employment Status: Employed

Occupation Type: Salaried

Residential Status: Own

City / Town: Urban

Co-Applicant: No

Financial Details **Loan Details**

Step 3 — Fill in Employment Details

The second section asks about your employment situation:

Field	What to Enter	Example
Employment Status	Whether you are currently Employed, Self-Employed, or Unemployed	Employed
Occupation Type	The type of work you do	Salaried
Residential Status	Whether you Own your home, Rent it, or have Other arrangements	Own
City / Town	Whether you live in an Urban, Suburban, or Rural area	Urban
Co-Applicant	Whether someone else is applying for the loan with you	No

Step 4 — Fill in Financial Details

The third section collects your financial information. This is the most important section as it has the greatest impact on the prediction.

Field	What to Enter	Example
Annual Income	Your total yearly income before taxes	85000

	(in your local currency)	
Monthly Expenses	Your average monthly spending on all expenses	2500
Credit Score	Your credit score from your bank or credit agency (300 to 850)	720
Existing Loans	How many loans you currently have open	1
Total Existing Loan Amount	The total amount of money you still owe on existing loans	15000
Outstanding Debt	Total unpaid debt across all accounts	8000
Loan History	Select whether you have had a previous loan	No Previous Loan
Bank Account History	How many years your bank account has been active	5
Transaction Frequency	Your average number of bank transactions per month	15

⚠ Important: Enter numbers only — do not include commas, currency symbols, or spaces. For example enter 85000 not 85,000 or \$85,000.

Step 5 — Fill in Loan Details

The fourth section asks about the loan you are applying for:

Field	What to Enter	Example
Loan Amount Requested	The total amount of money you want to borrow	20000
Loan Term	How many months you want to repay the loan over	120
Loan Purpose	What you will use the loan for	Home
Interest Rate (%)	The expected interest rate percentage	8.5
Loan Type	Secured means backed by an asset (e.g. property). Unsecured means no asset backing.	Secured

The screenshot shows a web browser window with the URL 'localhost:3002/predict'. A 'Screenshot captured' notification is visible. The form is titled 'FinPredict - Intelligent L...' and is divided into two main sections: 'Financial Details' and 'Loan Details'. The 'Financial Details' section includes input fields for Annual Income (e.g. 85000), Monthly Expenses (e.g. 2500), Credit Score (300-850) (e.g. 720), Existing Loans (e.g. 1), Total Existing Loan Amount (e.g. 15000), Outstanding Debt (e.g. 8000), Loan History (No Previous Loan), Bank Account History (years) (e.g. 5), and Transaction Frequency (e.g. 15). The 'Loan Details' section includes input fields for Loan Amount Requested (e.g. 20000), Loan Term (months) (e.g. 120), Loan Purpose (Home), Interest Rate (%) (e.g. 8.5), and Loan Type (Secured). A blue button at the bottom of the form reads 'Check Loan Eligibility'.

Step 6 — Submit Your Application

Once all fields are filled in, scroll to the bottom of the form and click the blue Check Loan Eligibility button.

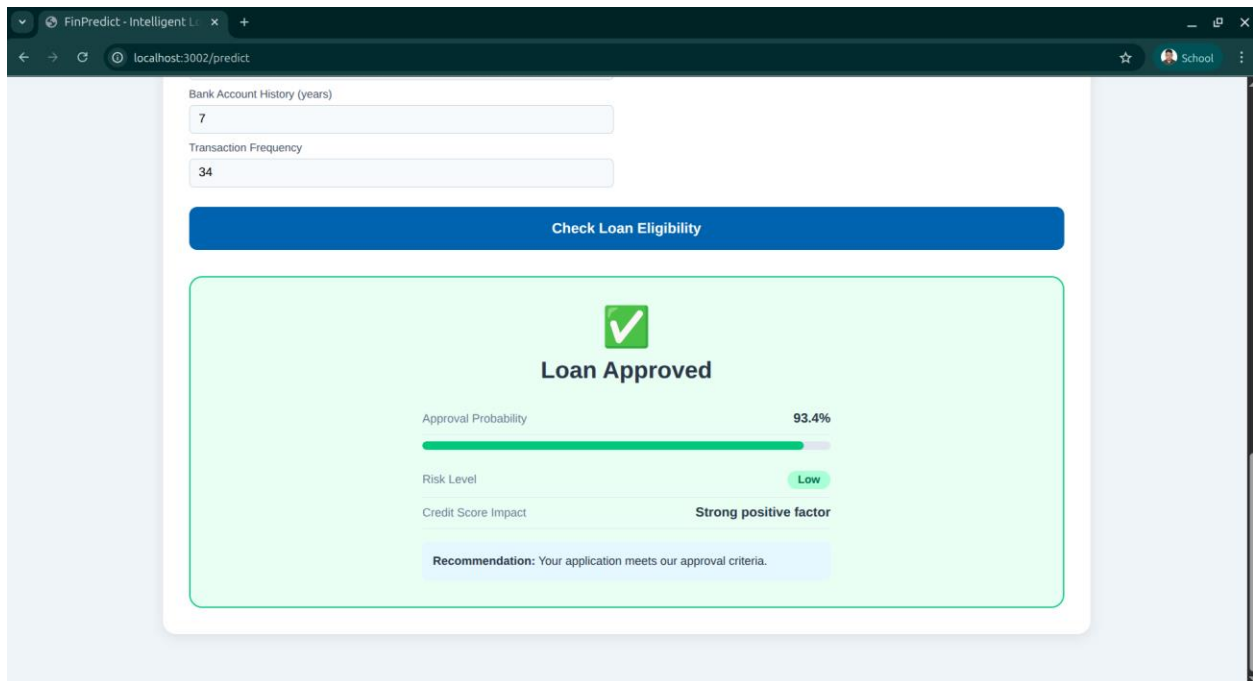
Tip: Make sure all fields are filled in before clicking the button. Empty fields will cause an error message to appear.

4. Understanding Your Result

After clicking Check Loan Eligibility, your result appears below the form within seconds. There are two possible outcomes:

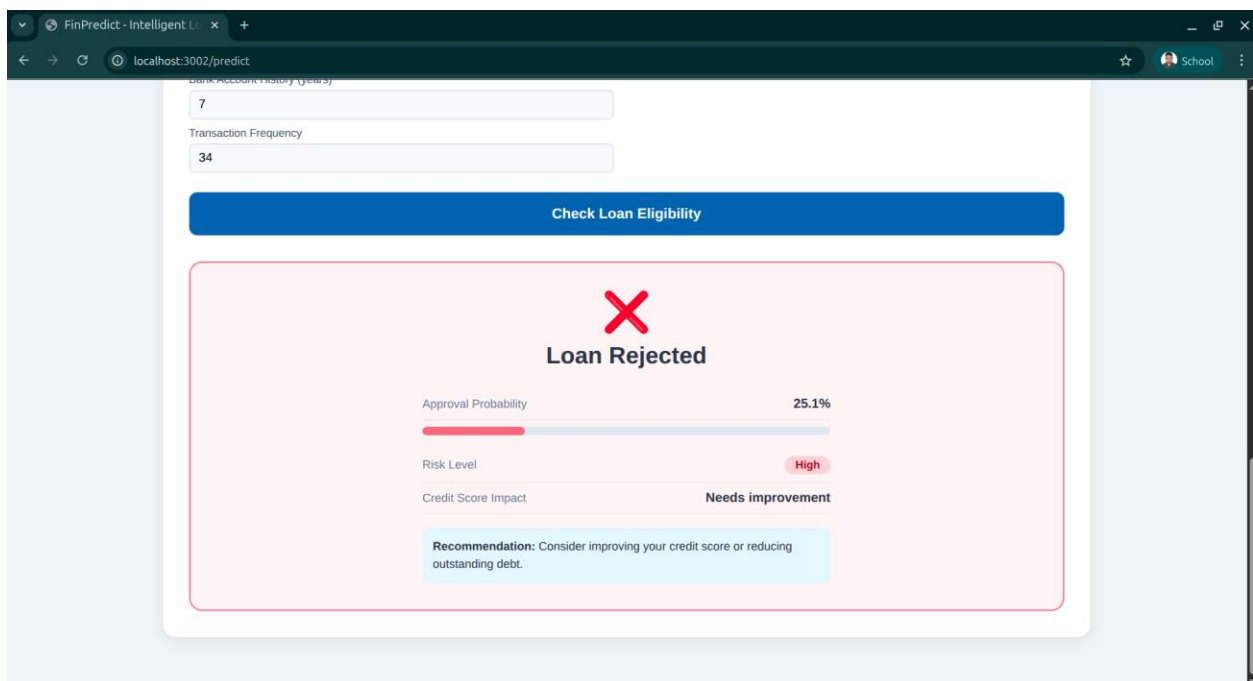
4.1 Loan Approved

A green result card appears with a checkmark icon and the message Loan Approved. This means your financial profile meets the criteria and your loan is likely to be approved.



4.2 Loan Rejected

A red result card appears with a cross icon and the message Loan Rejected. This means your current financial profile does not meet the approval criteria. This is a prediction to help you prepare — see Section 5 for tips on improving your chances.



4.3 What Each Field Means

Field	What It Means	Example
Approval Probability	A percentage showing how confident the AI is in the approval. Higher is better.	78.1% means strong chance of approval
Probability Bar	A visual bar showing the approval probability. Longer green bar = stronger chance.	80% filled bar = good profile
Risk Level	A badge showing Low, Medium, or High risk classification of your profile.	Low risk = strong application
Credit Score Impact	Whether your credit score is helping or hurting your application.	Needs improvement = score below 650
Recommendation	A personalised tip to improve your loan approval chances.	Improve credit score or reduce debt

4.4 Risk Level Explained

Risk Level	Meaning	What to Do
Low	Your profile is strong. Approval is likely.	Proceed with your formal application.
Medium	Your profile is borderline. Approval is possible but not guaranteed.	Consider improving one or two factors before applying.
High	Your profile has significant risk factors. Rejection is more likely.	Work on improving your financial profile first. See Section 5.

5. Tips to Improve Your Approval Chances

If your result shows Loan Rejected or a Medium / High risk level, here are practical steps to improve your profile before re-applying:

5.1 Improve Your Credit Score

- Pay all existing bills and loan instalments on time
- Reduce your credit card balances
- Avoid applying for multiple loans at the same time
- Check your credit report for errors and dispute any inaccuracies

 **Note:** A credit score above 700 significantly improves your approval chances. The most important factor in the FinPredict model is your credit score.

5.2 Reduce Your Outstanding Debt

- Pay off as much existing debt as possible before applying
- Aim to keep your Debt-to-Income ratio below 30% (outstanding debt less than 30% of annual income)
- Consider consolidating multiple debts into one lower-interest loan

5.3 Increase Your Income or Reduce Expenses

- Demonstrate a stable employment history
- Reduce your monthly expenses to lower your expense-to-income ratio
- Additional income sources (part-time work, rental income) can strengthen your application

5.4 Adjust Your Loan Request

- Request a smaller loan amount relative to your annual income
- Choose a longer loan term to lower monthly repayment amounts
- Consider applying for a Secured loan (backed by an asset) rather than Unsecured

5.5 Add a Co-Applicant

- Adding a co-applicant with a strong credit profile can significantly improve your chances
- The co-applicant shares responsibility for the loan repayment

 **Tip:** After making improvements, return to FinPredict and re-run the eligibility check to see if your new profile scores better before formally applying to a financial institution.

6. Troubleshooting Common Issues

Problem	Likely Cause	Solution
Prediction failed error appears	The backend server is not running	Contact your system administrator to restart the backend service
Form fields lose focus while typing	Browser issue	Refresh the page and try again in Google Chrome
The page does not load	Wrong URL or server not started	Check the URL and ensure the application is running
Numbers entered show an error	Numbers entered with commas or symbols	Remove all commas, currency symbols, and spaces — enter numbers only
Result does not match expectation	Some fields may be incorrectly filled	Review all 24 fields carefully and resubmit
Page is very small on mobile	Mobile screen is narrow	For best results use a tablet or desktop computer

7. Complete Field Reference

This section lists all 24 fields in the loan application form with accepted values and guidance.

7.1 Personal Details

Field	Accepted Values	Notes
Gender	Male / Female	Select from dropdown
Age	18 to 100	Numbers only. Must be 18 or above.
Marital Status	Married / Single / Divorced	Select from dropdown
Dependents	0 to 10	Number of financially dependent people
Education	Graduate / High School / Postgraduate	Select highest completed level

7.2 Employment Details

Field	Accepted Values	Notes
Employment Status	Employed / Self-Employed / Unemployed	Select from dropdown
Occupation Type	Salaried / Professional /	Select from dropdown

	Freelancer / Business	
Residential Status	Own / Rent / Other	Select your housing situation
City / Town	Urban / Suburban / Rural	Select your area type
Co-Applicant	Yes / No	Whether someone else is applying with you

7.3 Financial Details

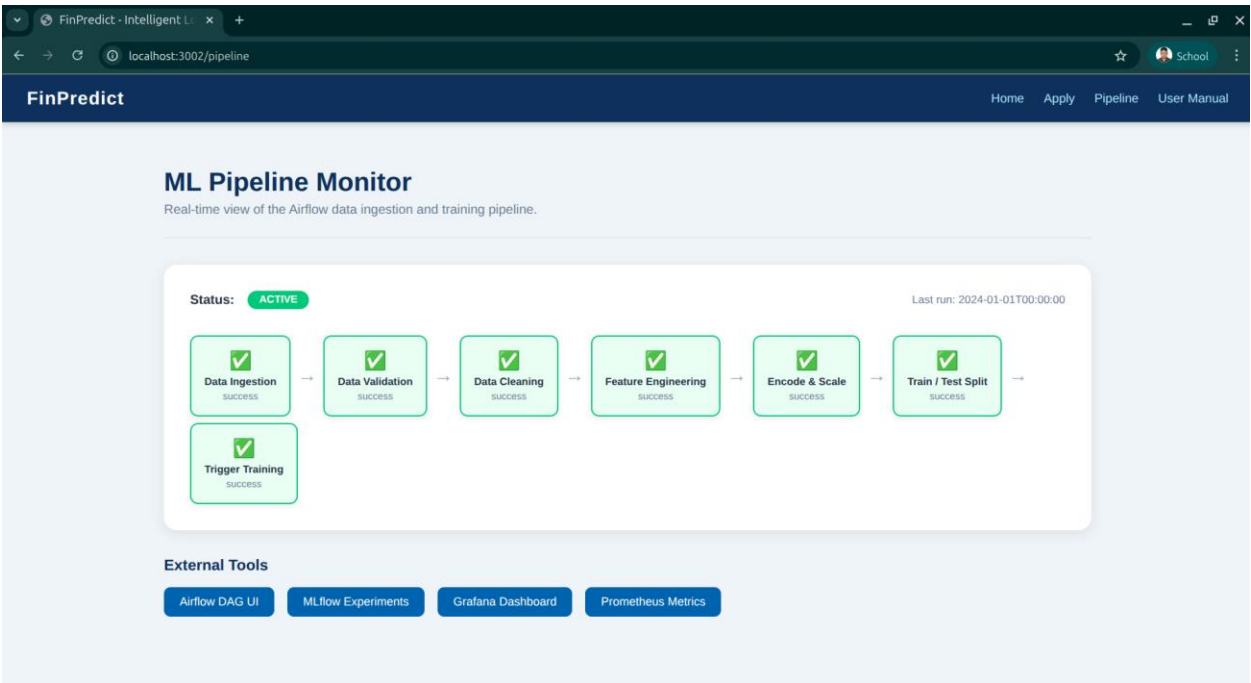
Field	Accepted Values	Notes
Annual Income	Any positive number	Total yearly income before tax. No commas or symbols.
Monthly Expenses	Any positive number	Average monthly spending. No commas or symbols.
Credit Score	300 to 850	From your bank or credit agency. Higher is better.
Existing Loans	0 or more	Number of currently open loans
Total Existing Loan Amount	0 or more	Total amount still owed on all existing loans
Outstanding Debt	0 or more	Total unpaid debt across all accounts
Loan History	No Previous Loan / Has Previous Loan	Select from dropdown
Bank Account History	0 or more	Number of years your account has been active
Transaction Frequency	0 or more	Average monthly bank transactions

7.4 Loan Details

Field	Accepted Values	Notes
Loan Amount Requested	Any positive number	Total amount you want to borrow. No commas or symbols.
Loan Term	Any positive number	Repayment period in months. E.g. 120 = 10 years.
Loan Purpose	Home / Personal / Vehicle / Education	Select from dropdown
Interest Rate (%)	Any positive number	Expected percentage. E.g. enter 8.5 for 8.5%
Loan Type	Secured / Unsecured	Secured = backed by an asset. Unsecured = no asset.

8. Pipeline Monitor (For Administrators)

The Pipeline page is designed for technical users and system administrators. It shows the real-time status of the data processing and AI training pipeline that powers FinPredict.



8.1 Task Status Indicators

Status	Meaning
success (green border)	This pipeline task completed successfully
failed (red border)	This pipeline task encountered an error — contact administrator

8.2 External Tool Links

The Pipeline page provides direct links to the following monitoring tools:

Tool	URL	Purpose
Airflow DAG UI	http://localhost:8080	View and manage the 7-task data pipeline
MLflow Experiments	http://localhost:5000	View model training experiments and metrics
Grafana Dashboard	http://localhost:3001	Real-time system monitoring dashboard
Prometheus Metrics	http://localhost:9090	Raw metrics and alert rules

9. Glossary

This section explains technical terms used in the application.

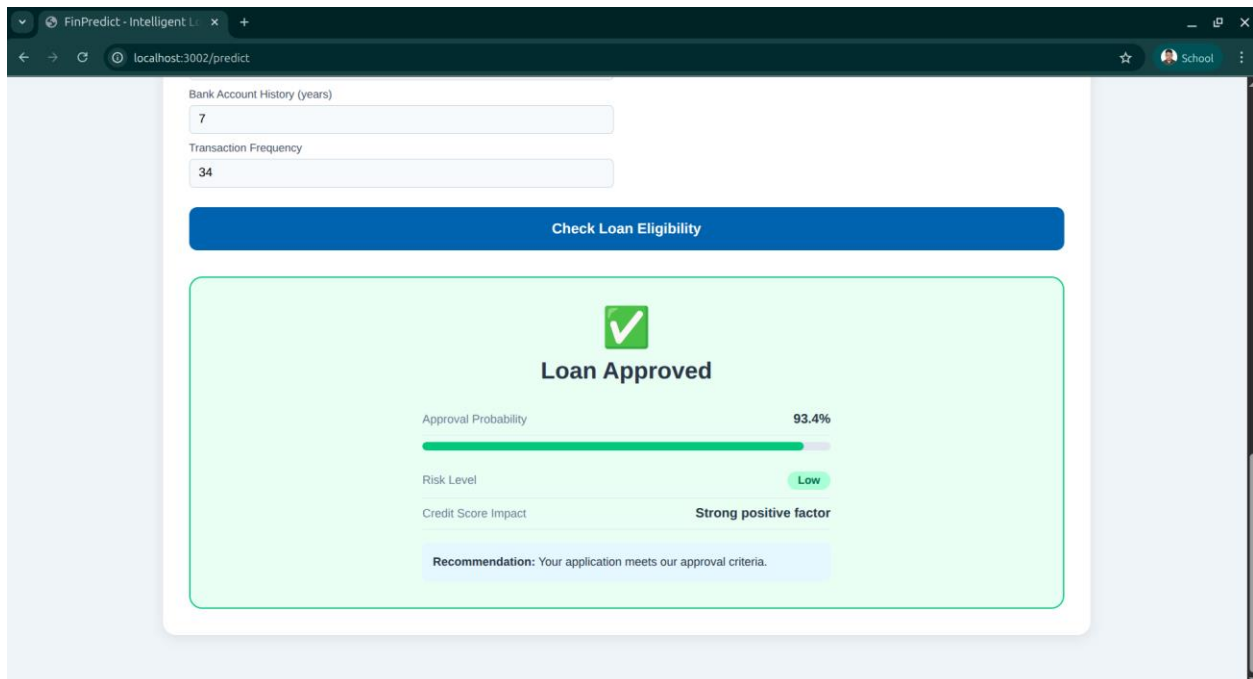
Term	Plain English Explanation
AI / Artificial Intelligence	A computer system that learns from data to make decisions, similar to how a human expert would
Approval Probability	A percentage showing how confident the system is that your loan will be approved
Credit Score	A number between 300 and 850 that represents your creditworthiness. Higher is better.
Debt-to-Income Ratio	Your total debt divided by your annual income. Lower is better for loan approval.
Secured Loan	A loan backed by an asset (e.g. your home or car) which the lender can claim if you do not repay
Unsecured Loan	A loan with no asset backing — higher risk for the lender, so usually higher interest rates
Co-Applicant	A second person who applies for the loan with you and shares repayment responsibility
Loan Term	The number of months over which you agree to repay the loan
Outstanding Debt	The total amount of money you currently owe across all accounts
Risk Level	A Low / Medium / High classification of how risky your loan profile appears to the AI model
Prediction	The AI system's best estimate of the loan outcome based on your data
Pipeline	The automated sequence of steps that processes data and trains the AI model

10. Quick Start Summary

Follow these 6 steps to check your loan eligibility:

1. Open your browser and go to <http://localhost:3002>
2. Click Apply in the navigation bar
3. Fill in Personal Details — Gender, Age, Marital Status, Dependents, Education
4. Fill in Employment Details — Employment Status, Occupation Type, Residential Status, City, Co-Applicant
5. Fill in Financial Details — Income, Expenses, Credit Score, Loans, Debt
6. Fill in Loan Details — Amount, Term, Purpose, Interest Rate, Type
7. Click the blue Check Loan Eligibility button
8. Read your result — Approved or Rejected — and follow the recommendation

✓ **Done!** If you see Loan Approved with a high probability and Low risk, your application is looking strong. Consider formally applying to your financial institution.



The screenshot shows a web browser window with the address bar displaying 'localhost:3002/predict'. The page contains a form with the following fields:

- Bank Account History (years): 7
- Transaction Frequency: 34

Below the form is a blue button labeled 'Check Loan Eligibility'. The result is displayed in a green box with a large green checkmark and the text 'Loan Approved'. The following metrics are shown:

- Approval Probability: 93.4% (represented by a green progress bar)
- Risk Level: Low (represented by a green pill)
- Credit Score Impact: Strong positive factor

A recommendation box at the bottom states: 'Recommendation: Your application meets our approval criteria.'